

Analytical techniques

A Level Chemistry

Thin-layer chromatography

Chromatography 色谱 separates a mixture using two "phases" —one that stays still and one that moves. In **thin-layer chromatography** 薄层色谱 (TLC):

- the **stationary phase** 固定相 stays still (for example aluminium oxide on a plate).
- the **mobile phase** 流动相 moves (a polar or non-polar solvent that travels up the plate).
- the **baseline** 基线 is the starting line where the spots are placed; the **solvent front** 溶剂前沿 is the highest level the solvent reaches.

The R_f value compares how far a spot moves with how far the solvent moves:

$$R_f = \frac{\text{distance moved by the spot}}{\text{distance moved by the solvent front}}$$

It is always between 0 and 1. A substance that sticks more strongly to the stationary phase, or is less soluble in the mobile phase, moves less and has a **smaller** R_f .

Gas/liquid chromatography

In **gas/liquid chromatography** 气液色谱 (GLC):

- the stationary phase is a high-boiling-point non-polar liquid on a solid support.
- the mobile phase is an unreactive carrier gas.
- the **retention time** 保留时间 is how long a component takes to pass through.

The area of each peak gives the percentage of that component in the mixture. A component that interacts more with the stationary phase has a longer retention time.

Carbon-13 NMR spectroscopy

NMR 核磁共振 (nuclear magnetic resonance) studies how certain nuclei behave in a strong magnetic field.

In carbon-13 NMR, each different **chemical environment** 化学环境 of carbon gives one peak. So:

- the **number of peaks** tells you how many different carbon environments there are (equivalent carbons share a peak).
- the position of each peak (its chemical shift) suggests the type of carbon, which helps you deduce possible structures.

Proton (^1H) NMR spectroscopy

Proton NMR looks at the hydrogen atoms (**proton** 质子 nuclei). From the spectrum you read off:

- **chemical environments:** protons in different environments appear at different **chemical shift** 化学位移 values.
- **relative numbers:** the relative peak **areas** give the ratio of each type of proton.
- **splitting** 裂分: a peak is split by the protons on the neighbouring carbon, following the **$n + 1$ rule** — n equivalent neighbours split a peak into $n + 1$ lines:

Neighbours (n)	Pattern
0	singlet 单峰
1	doublet 双峰
2	triplet 三峰
3	quartet 四峰
many	multiplet 多重峰

Practical points

- **tetramethylsilane** 四甲基硅烷 (TMS) is the standard, set at a chemical shift of 0.
- a **deuterated solvent** 氘代溶剂 (such as CDCl_3) is used so that the solvent itself gives no proton signal.
- shaking the sample with D_2O makes the O–H and N–H peaks disappear (their hydrogen is swapped for deuterium), which identifies those protons.